EXPERIMENT 6

Aim: Creation and Deployment of Smart Contract and Transactions using Solidity and Remix IDE**.**

THEORY

1. Smart Contract

A smart contract is a program deployed on a blockchain that automatically executes predefined rules when its functions are called. In this experiment, a Ballot/Voting smart contract is created using Solidity and deployed using Remix IDE.

The Ballot contract allows:

- The chairperson to grant voting rights to voters.
- Voters to cast their votes.
- Voters to delegate their votes to other voters.
- The contract to calculate the winning proposal.
- The name of the winning proposal to be displayed.

2. Require Statement

The `require` statement is used as a guard condition in Solidity. It checks whether a particular condition is true before allowing a function to execute.

If the condition is false, the function execution stops and the transaction is reverted. Any changes made during that transaction are rolled back.

It is used to ensure that only authorized users can perform operations and to prevent invalid transactions. Thus, it improves the security and reliability of smart contracts.

3. Mapping

A mapping is a Solidity data structure that associates keys with values, similar to a hash table. In the Ballot contract, mapping connects Ethereum addresses with their corresponding voter information.

The voter information includes voting weight, voting status, delegated address and proposal index. Mappings provide efficient data lookup but do not have a length property and cannot be directly iterated over.

4. Storage

Storage refers to permanent data stored on the blockchain. Variables stored in storage remain available across transactions until they are modified.

In the Ballot contract, voter information is maintained in storage throughout the contract's lifecycle. However, storing data on the blockchain consumes gas, making storage more expensive.

5. Memory

Memory is temporary data storage used during function execution. It is mainly used for temporary variables, function arguments and computations.

The data stored in memory is discarded once the function execution is completed. Compared to storage, memory is less expensive and does not permanently store information on the blockchain.

6. Why bytes32 Instead of string?

`bytes32` is a fixed-size data type that stores exactly 32 bytes of data. It provides efficient storage, faster comparisons and lower gas consumption. However, it limits the proposal name to 32 bytes.

On the other hand, `string` is a dynamically sized data type that supports variable-length text. It is more flexible and readable but may require more gas and complex handling.

Therefore, `bytes32` is preferred when gas efficiency is important, whereas `string` is preferred for flexibility and readability.

7. Ballot Contract

The Ballot contract implements a voting mechanism using Solidity. It manages voters, voting rights, vote delegation and proposal selection. It also calculates the winning proposal based on the total votes received and displays the winner's name.

Output & Code:

Structure of Voter and Proposal

```
struct Voter {
    uint weight;
    bool voted;
    address delegate;
    uint vote;
}

struct Proposal {
    string name;
    uint voteCount;
}
```

Constructor

```
constructor(string[] memory proposalNames) {
    chairperson = msg.sender;
    voters[chairperson].weight = 1;

    for (uint i = 0; i < proposalNames.length; i++) {
        proposals.push(Proposal({
```

```
            name: proposalNames[i],
            voteCount: 0
        }));
    }
}
```

Give Right to Vote

```
function giveRightToVote(address voter) external {
    require(msg.sender == chairperson,
        "Only chairperson can give right to vote.");

    require(!voters[voter].voted,
        "The voter already voted.");

    require(voters[voter].weight == 0,
        "Voter already has the right to vote.");

    voters[voter].weight = 1;
}
```

Vote Delegation

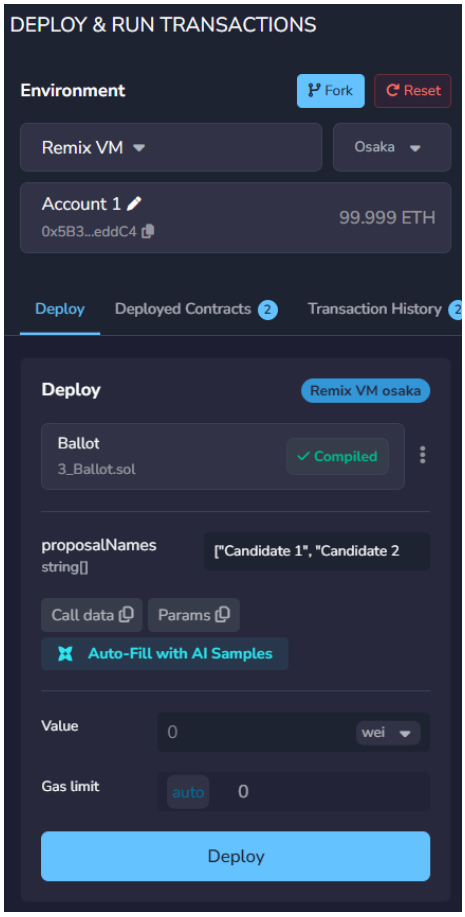
```
function delegate(address to) external {
    Voter storage sender = voters[msg.sender];

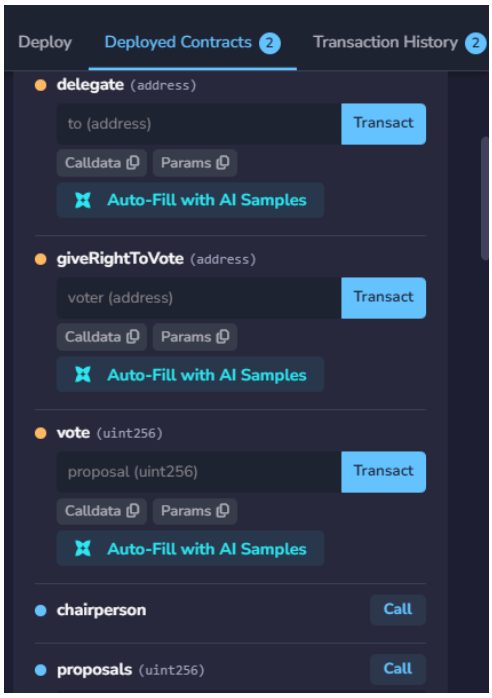
    require(sender.weight != 0, "You have no right to vote");
    require(!sender.voted, "You already voted.");
    require(to != msg.sender, "Self-delegation is disallowed.");

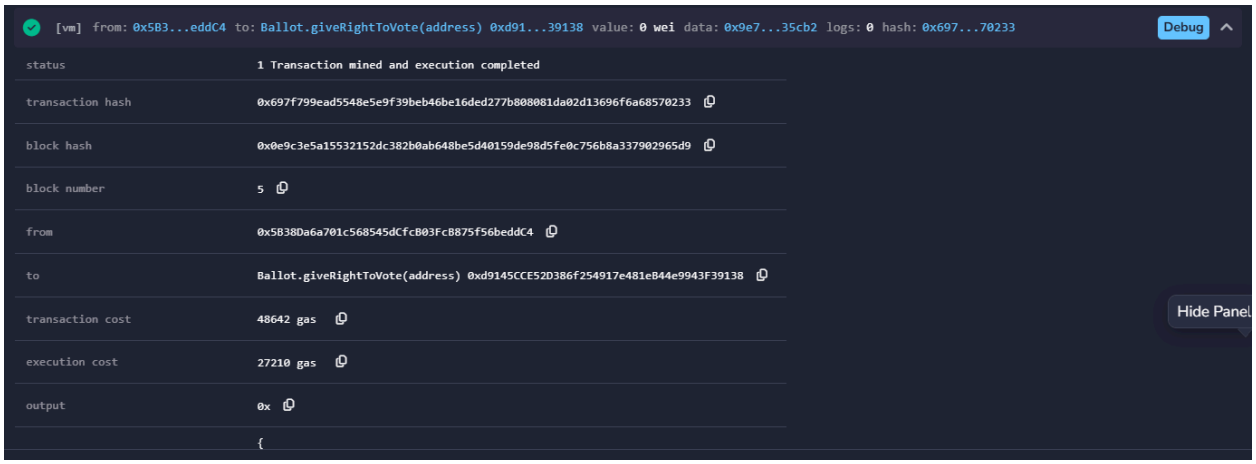
    Voter storage delegate_ = voters[to];
    require(delegate_.weight >= 1);

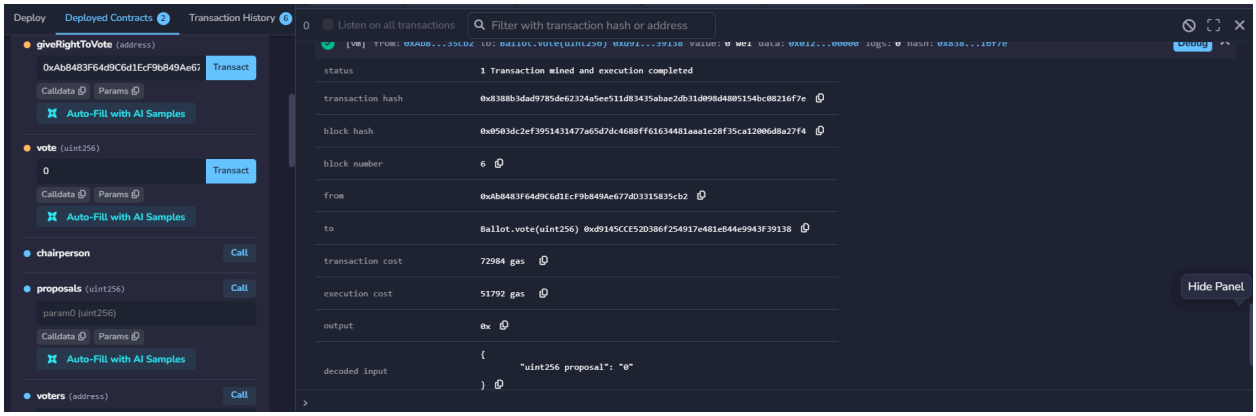
    sender.voted = true;
    sender.delegate = to;

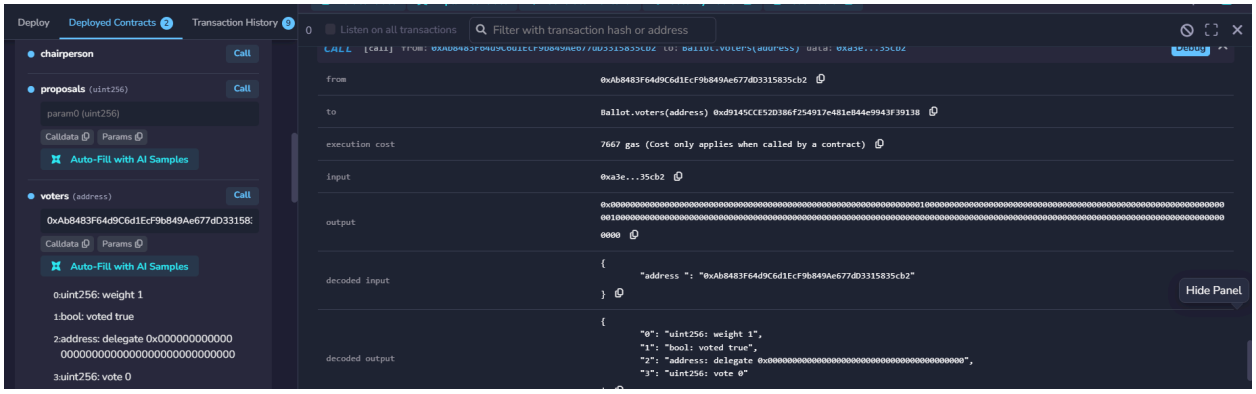
    if (delegate_.voted) {
        proposals[delegate_.vote].voteCount += sender.weight;
    } else {
        delegate_.weight += sender.weight;
    }
}
```


DEPLOY & RUN TRANSACTIONS
Environment
Fork
Reset
Remix VM
Osaka
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed Contracts 2
Transaction History 2
Deploy
Remix VM osaka
Ballot
3_Ballot.sol
Compiled
proposalNames
string[]
["Candidate 1", "Candidate 2
Call data
Params
Auto-Fill with AI Samples
Value
0
wei
Gas limit
auto
0
Deploy


Deploy
Deployed Contracts 2
Transaction History 2
delegate (address)
to (address)
Transact
Calldata
Params
Auto-Fill with AI Samples
giveRightToVote (address)
voter (address)
Transact
Calldata
Params
Auto-Fill with AI Samples
vote (uint256)
proposal (uint256)
Transact
Calldata
Params
Auto-Fill with AI Samples
chairperson
Call
proposals (uint256)
Call


[vm] from: 0x5B3...eddC4 to: Ballot.giveRightToVote(address) 0xd91...39138 value: 0 wei data: 0x9e7...35cb2 logs: 0 hash: 0x697...70233
Debug
status
1 Transaction mined and execution completed
transaction hash
0x697f799ead5548e5e9f39beb46be16ded277b808081da02d13696f6a68570233
block hash
0x0e9c3e5a15532152dc382b0ab648be5d40159de98d5fe0c756b8a337902965d9
block number
5
from
0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to
Ballot.giveRightToVote(address) 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost
48642 gas
execution cost
27210 gas
output
0x
{
Hide Panel





Conclusion

Successfully developed and deployed an ETH Transfer smart contract using Solidity on Remix IDE. MetaMask was integrated to perform ETH transactions on the Sepolia Testnet. The transfer was successfully executed and verified using Sepolia Etherscan. Thus, the working of smart contracts and blockchain transactions was understood.